

A Computational Examination of Virtue Ethics Presence in Three Textual Datasets

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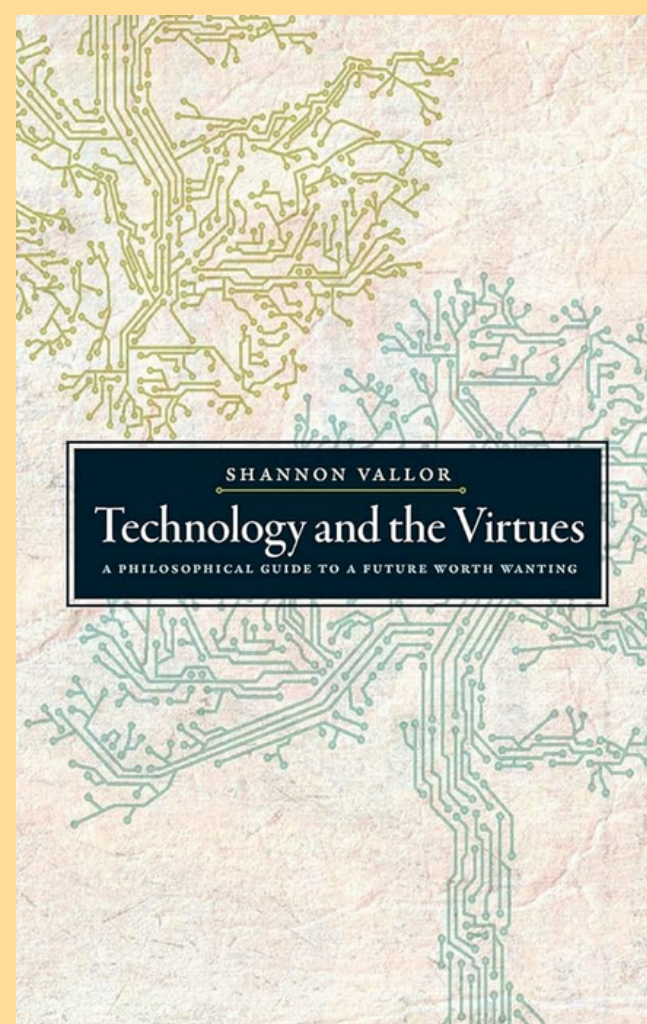


Background

Ethics of technology is a critical area of study as its impacts are reflected on both individuals and societies through the use of technologies. More broadly, people have enriched their lives and communities by following ethical principles. In her book *Technology and the Virtues*, Shannon Vallor outlined 12 techno-moral virtues and talked about the importance of practicing those virtues as both users and builders of technologies. Our research asks this question: **to what extent can we observe evidence of presence or engagement with Vallor's virtues in the textual sources of Wellesley's curriculum, the Wellesley News, and the and personal communications on the anonymous messaging board Sidechat?** These sources were then scored and analyzed by both humans and GPT. The validation of the GPT analysis can be found in the companion poster, *An Assessment of GPT-Prompting for Labeling Text in the Context of Virtue Ethics*.

The Technomoral Virtues

1. **Honesty**
2. **Self-control**
3. **Humility**
4. **Justice**
5. **Courage**
6. **Empathy**
7. **Care**
8. **Civility**
9. **Flexibility**
10. **Perspective**
11. **Magnanimity**
12. **Technomoral wisdom**



Shannon Vallor, *Technology and the Virtues*

Summary of the Datasets

Course Catalog: 1815 courses were collected from the Wellesley Course catalog. They were **evaluated for the extent they touch upon Shannon Vallor's technomoral virtues**. A score of 5 indicates that the virtue is minimally present and a score of 1 indicates that the virtue is very significantly present.

Wellesley News: 3504 articles from 2013 to 2022 were collected and then measured on a 5-point scale for the their **relevance to the dimensions of "technology and its use" and "ethics and societal impact"**. A score of 1 means "minimally mentions", and a score of 5 means "heavily mentions".

SideChat: 2265 posts were collected and **measured** on a 3-point scale, which was used to measure two dimensions, **Tone** and **Content**. The Tone is scored from -1 to 1, where -1 means being **rude** and 1 means being **polite**. The Content also uses the same scale, where -1 means the **opposite of Shannon Vallor's technomoral virtues** and 1 means **displaying the virtue fully**. 0 means neutral for both dimensions.

Wellesley Course Catalog

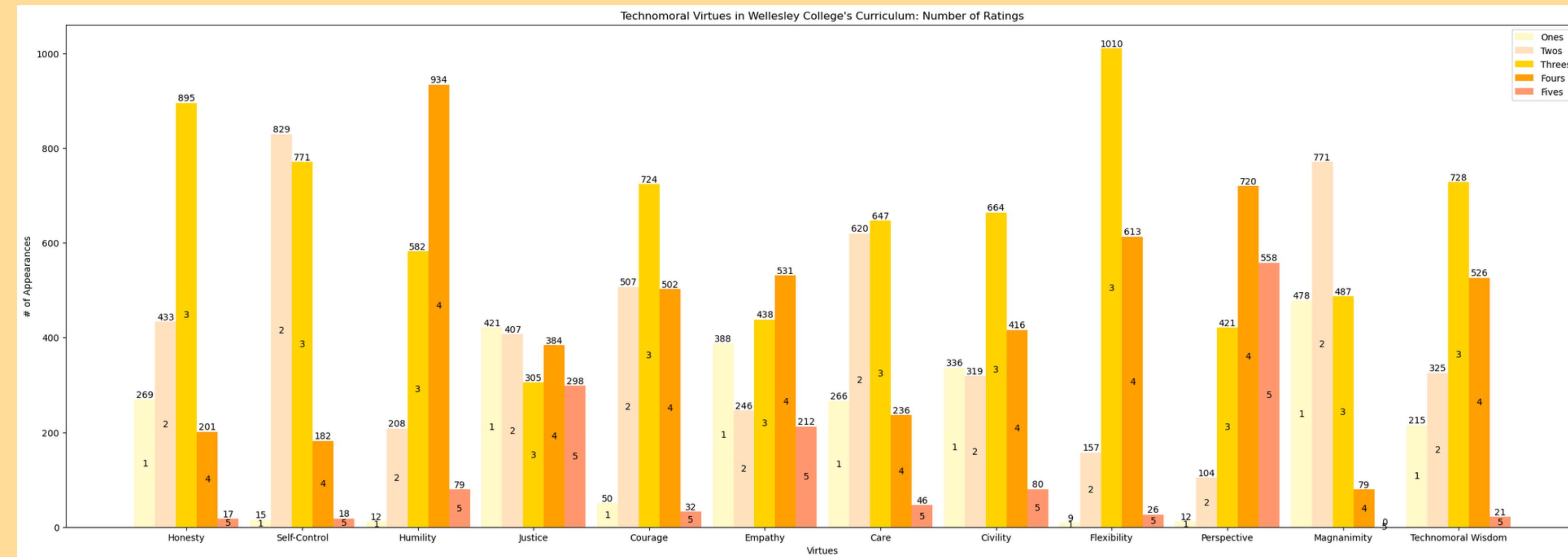


Figure 1. Bargraph for the distribution of scores according to each virtue

Perspective (558), **Justice (298)**, and **Empathy (212)** are the virtues that most frequently scored a five. **Magnanimity (0)**, **Honesty (17)**, and **Self-Control (18)** were the virtues that least frequently scored a five

708 (40%) Courses that recieved at least one score of 5

390 (21%) Courses that recieved at least two scores of 5

2.92 The mean score (the mean across all virtues in a course)

8 The greatest number of virtues scoring a 5 in one course. The one course that had a score of five on 8 virtues was PEAC / POL3 392: Truth Commissions and Conceptual Studies.

Wellesley News

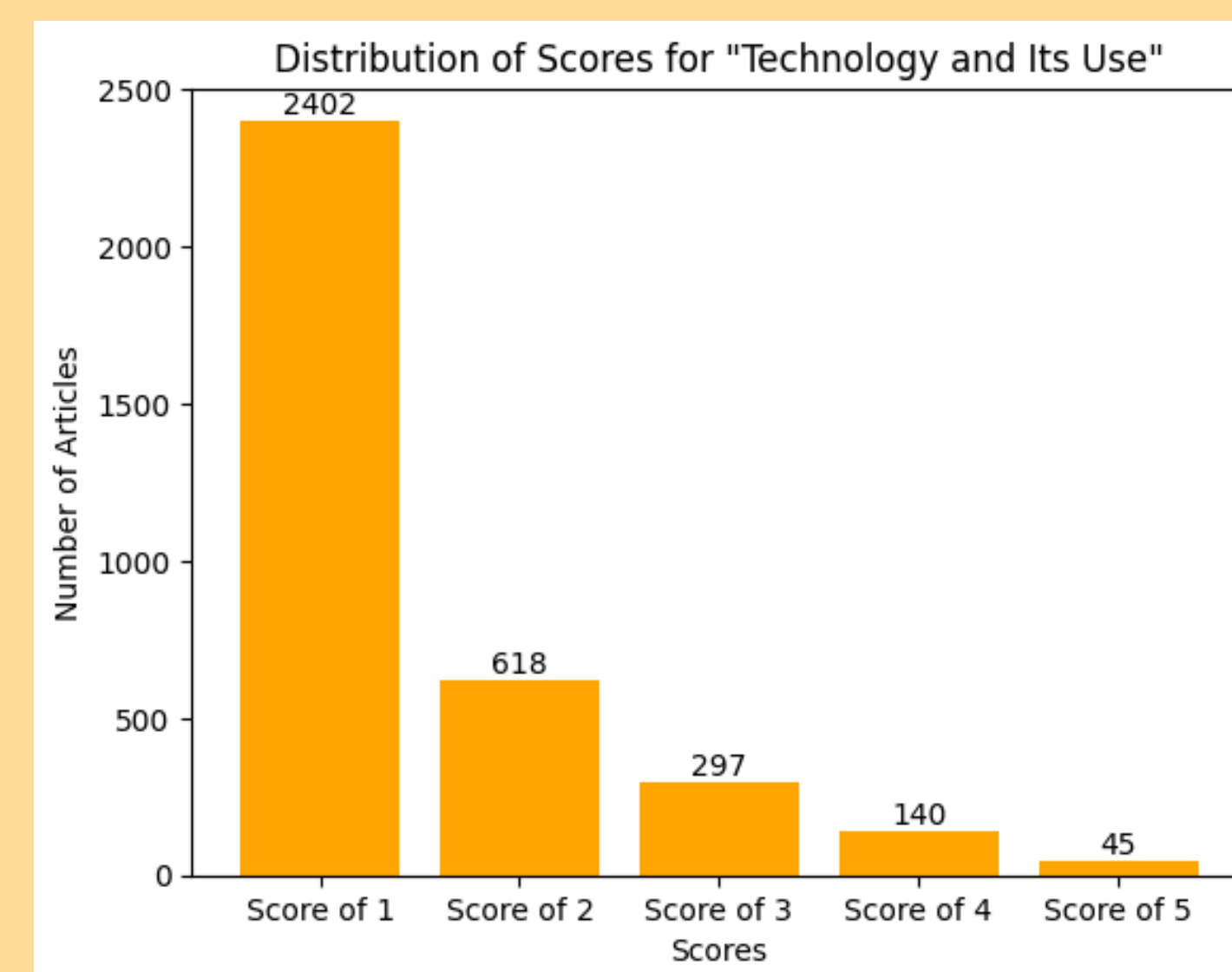


Figure 2. Bargraph for the distribution of scores for "Technology and Its Use"

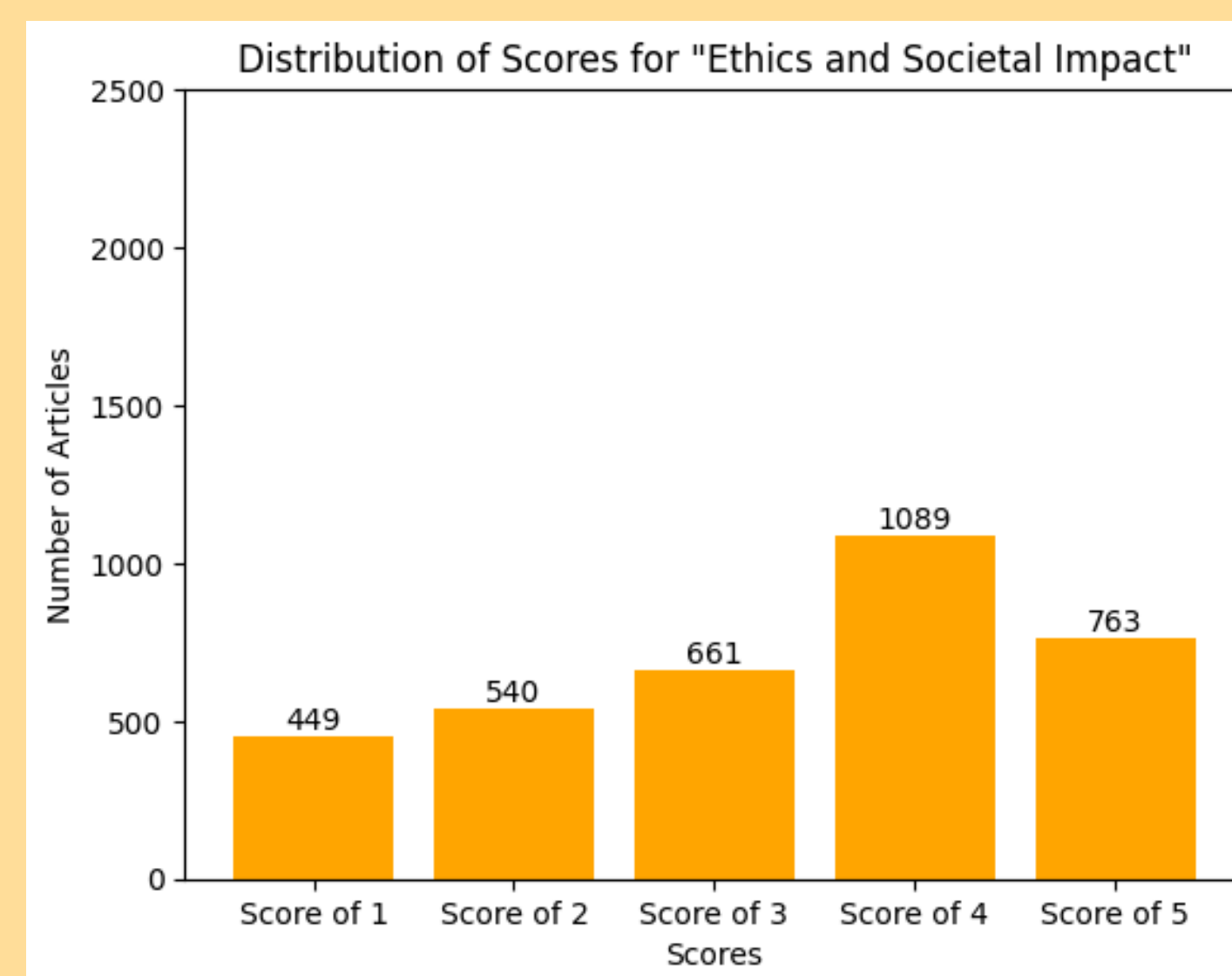


Figure 3. Bargraph for the distribution of scores for "ethics and societal impact"

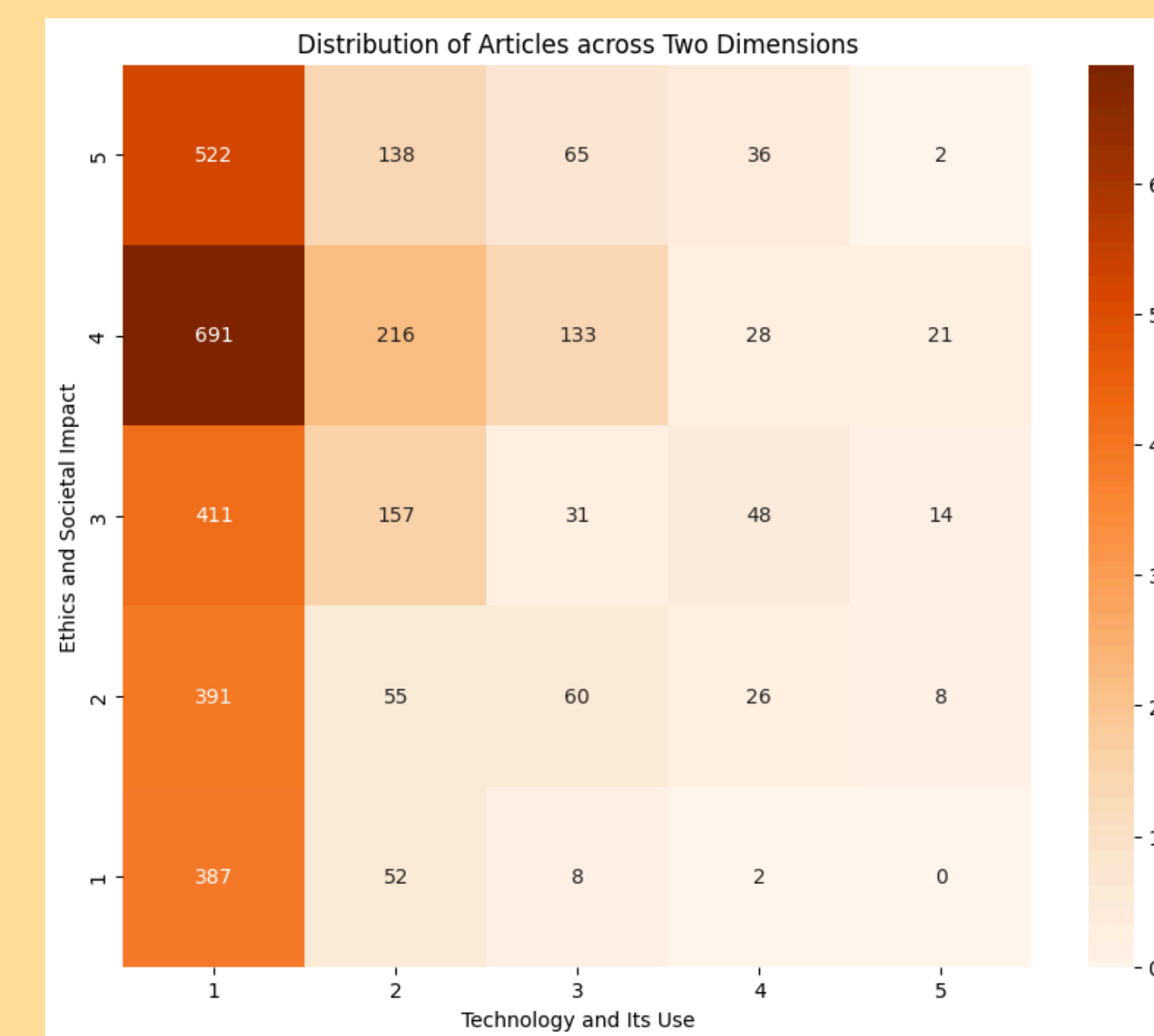


Figure 4. Heatmap for the distribution of articles across the two dimensions "technology and its use" and "ethics and societal impact"

SideChat

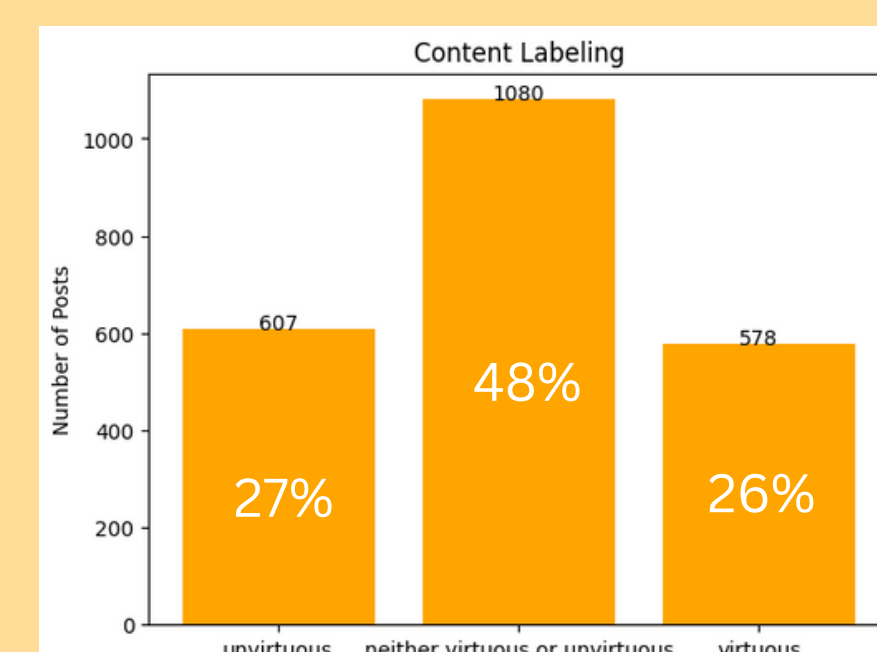


Figure 5, Bar graph showing how gpt labeled the content of posts within the sample set of 2265 posts

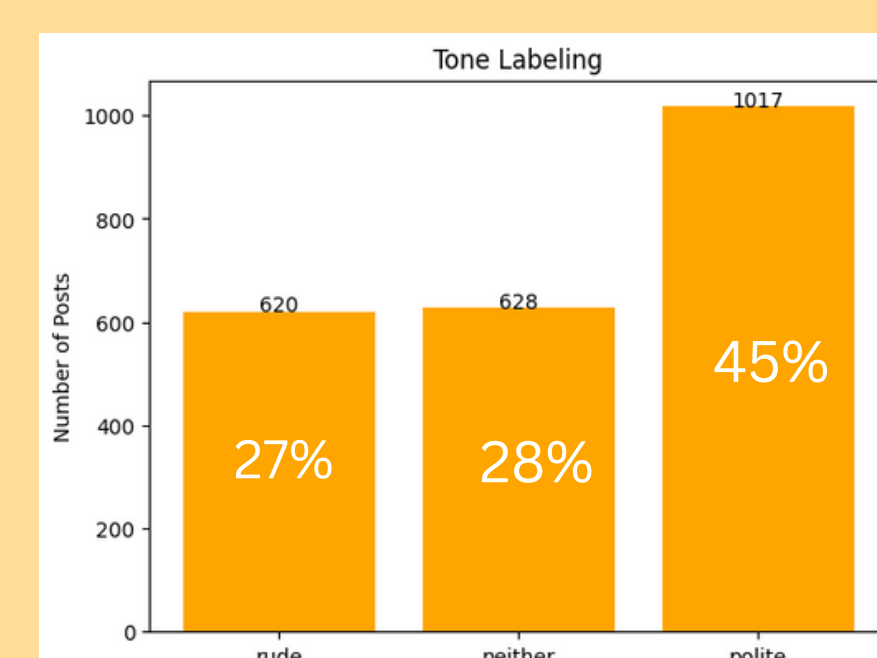


Figure 6. Bar graph showing how gpt labeled the tone of posts within the sample set of 2265 posts

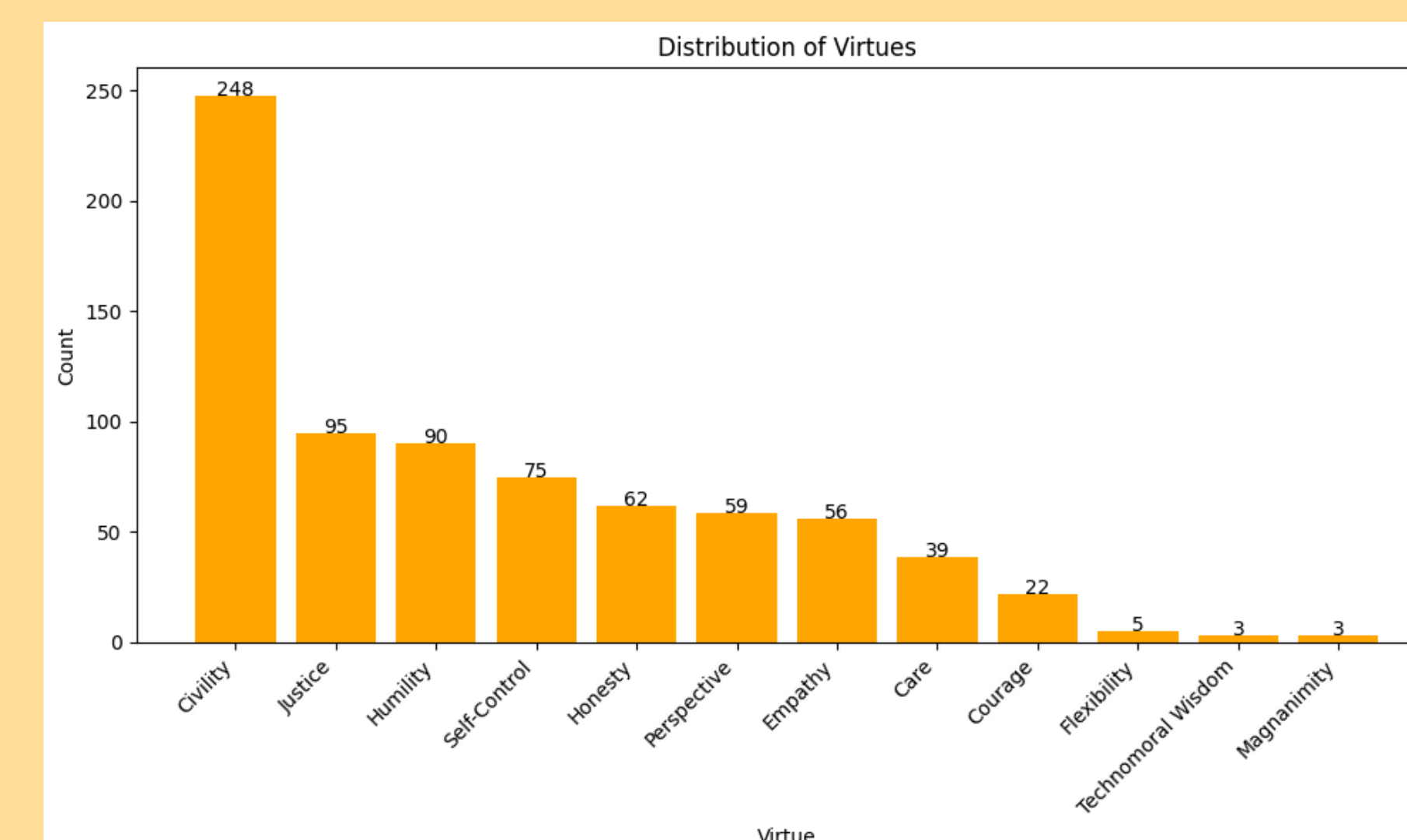


Figure 7. Bar graph of the distribution of the strongest virtue present in each post in a sample set of 2265 posts

Labeled Example:

Message: El Table workers are the best 🙌🙌🙌🙌

Tone: 1
Content: 1
Strongest Virtue: Civility

Tone Score Reasoning:
The message is very positive and uplifting, using praise and friendly emojis.

Content Score Reasoning:
It promotes a positive social interaction by showing appreciation and respect for others' work.

Discussion

Course Catalog: The Wellesley course curriculum displays many virtues, particularly perspective, justice, and empathy. Very few courses have no significant emphasis on at least one virtue, with the vast majority of courses somewhat focusing on at least one technomoral virtue (scores of 3 and 4). **65 courses** (3%) exhibited at least 4 virtues, with **17 (0.93)** courses exhibiting at least 5.

Wellesley News: Having a score of 4 or 5 was defined as having high relevance to the dimension the score measures. 5.28% (185/3504) of the articles collected received a score of high relevance to technology and its use and 52.85% (1852/3504) of the articles collected received a score of high relevance for ethics and societal. Only 2.48% (87/3504) of the articles received a score of high relevance for both dimensions, meaning that only a small number of them extensively discuss ethics and societal impact under technological contexts.

Sidechat: The **3 most prevalent virtues** were **civility** with 248 (11%) posts, **justice** with 95 (4%) posts, and **humility** with 90 (4%) posts. The number of virtuous posts GPT labeled is quite surprising. Despite the rhetoric that Sidechat is a toxic place, the data shows less than 30% of posts were labeled as unvirtuous or uncivil.

Overall, there is a presence of virtues within the three datasets was a pleasant surprise. If these virtues are to be enforced it could strengthen the relationship in the way individuals use technology, which Vallor's emphasizes.

Future Work

Course Catalog

- Interview students/professors about courses that focus on the virtues
- Consider how curricular and extracurricular activities provide opportunities for practicing these virtues
- Build an interactive website to provide an annotated exploration of the curriculum across the virtues

Wellesley News

- Include articles before 2013 and after 2022 into the dataset to analyze its changes regarding technological and ethical topics over time
- Encourage CS students to contribute news articles to Wellesley News that cover topics of ethics of technology and its dilemmas

Sidechat

- **Interview students** about their opinons on the influence of Sidechat conversations on their ethical decision-making
- Label the virtuous posts **based on topic** (ex: academics, politics, relationships, ect.)

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